

BirdCLEF+ 2026

Status Report

Wildlife species identification from passive acoustic monitoring
in the Pantanal wetlands (Brazil). Kaggle code competition.

Metric: macro ROC-AUC. Deadline: June 3, 2026.

Current state

Best LB: 0.897 (nb19a: Perch+MLP + top-K postproc)

Gap to 2025 winning solution (0.96): 0.063

CNN track: 5-fold ensemble trained, OOF macro-AUC 0.9792

(real LB pending submission; OOF likely overstates by 0.05-0.10)

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Dataset overview

Taxa breakdown

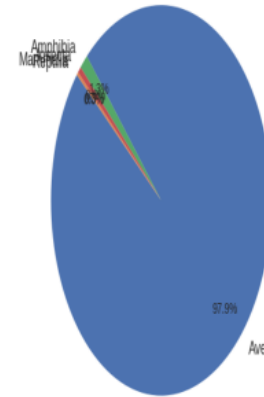
BirdCLEF+ 2026 -- key numbers

Total training clips	35,549
Species (multi-taxa)	234
Aves (birds)	162
Amphibia	35
Insecta	28 (+25 sonotypes)
Mammalia	8
Reptilia	1 (Caiman, 1 clip)

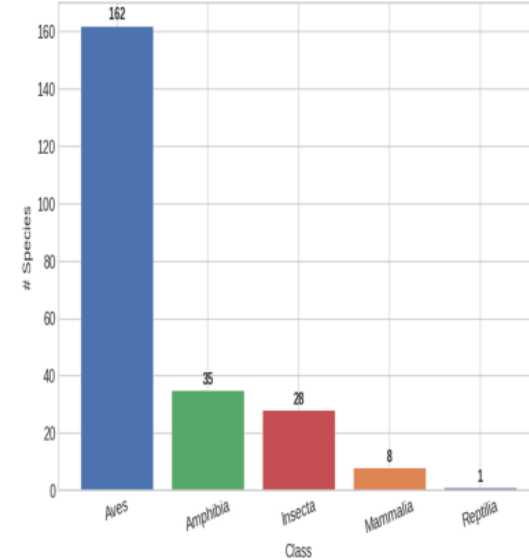
train_audio (XC + iNat)	35,549
train_soundscapes	10,658
Labeled segments	1,478 (66 files)

Region: Pantanal, Brazil
Lat -21.6 / -16.5, Lon -57.6 / -55.9
Format: OGG, 32 kHz

Training Clips by Taxonomic Class

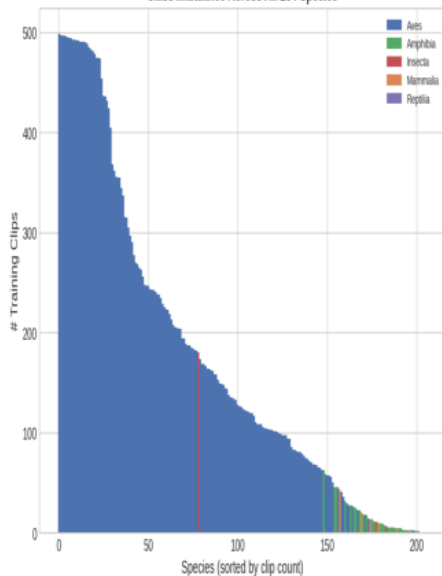


Number of Species per Class

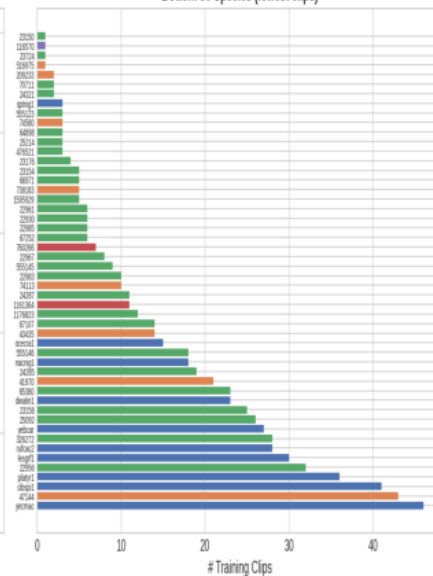


Class imbalance (long tail)

Class Imbalance Across All 234 Species

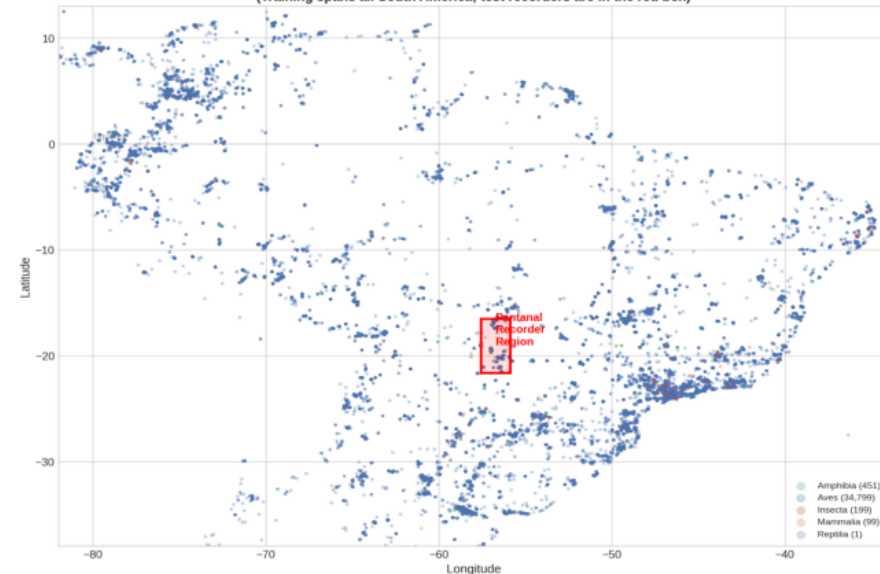


Bottom 50 Species (fewest clips)

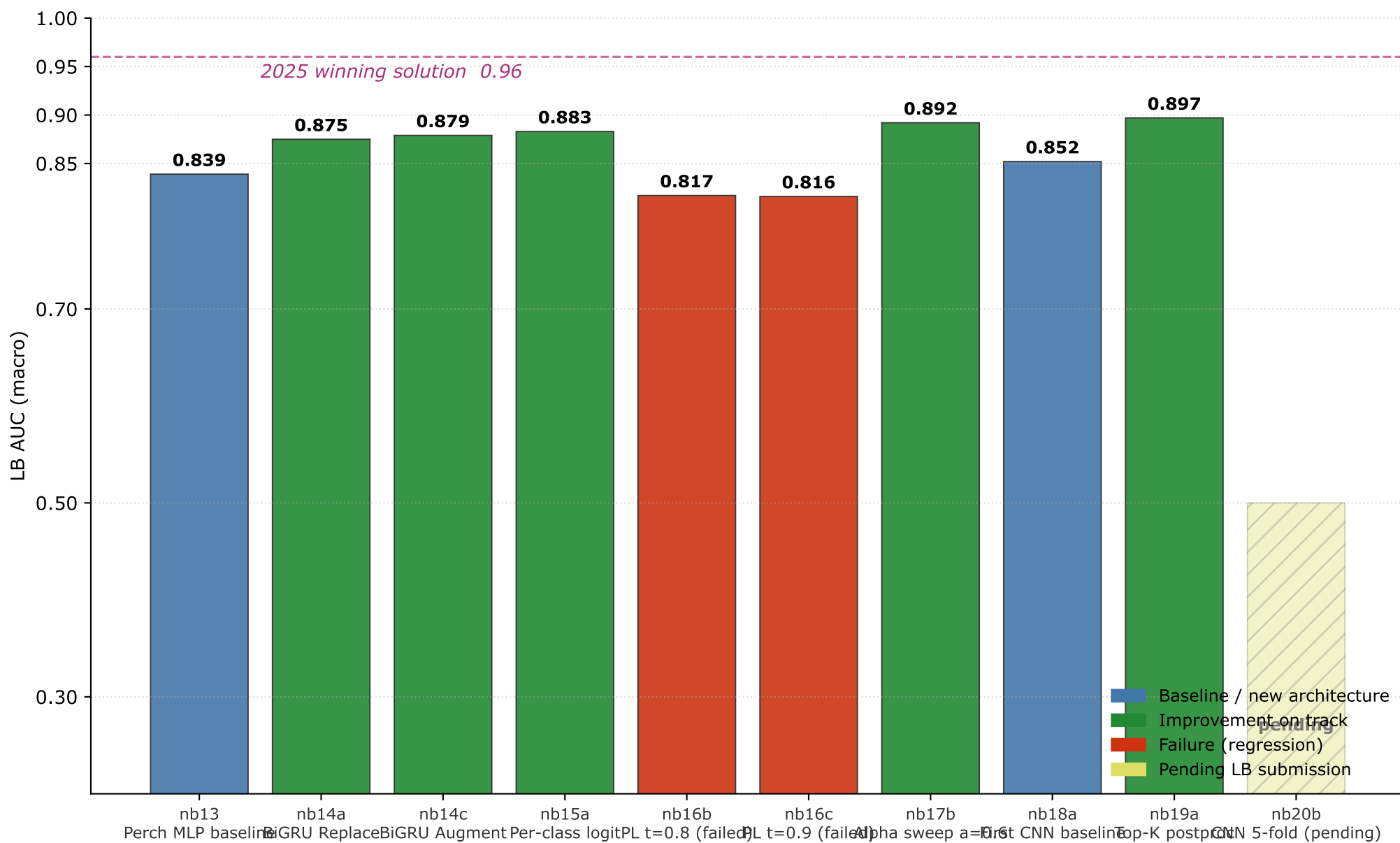


Geographic domain shift

Training Data Coverage vs Pantanal Recorder Region
(Training spans all South America; test recorders are in the red box)

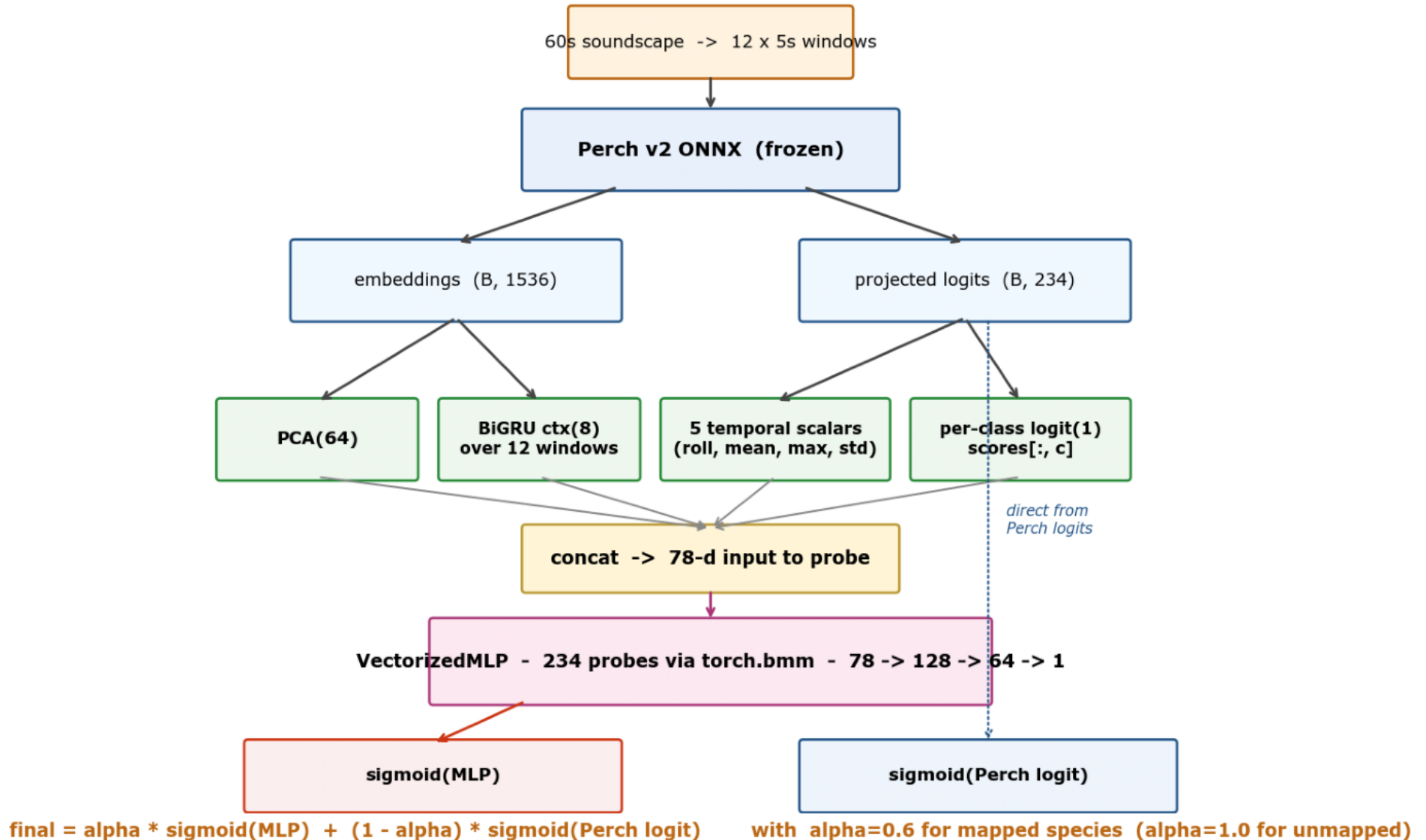


LB progression -- two tracks, ten notebooks



Perch+MLP architecture (LB 0.892 single, 0.897 with top-K)

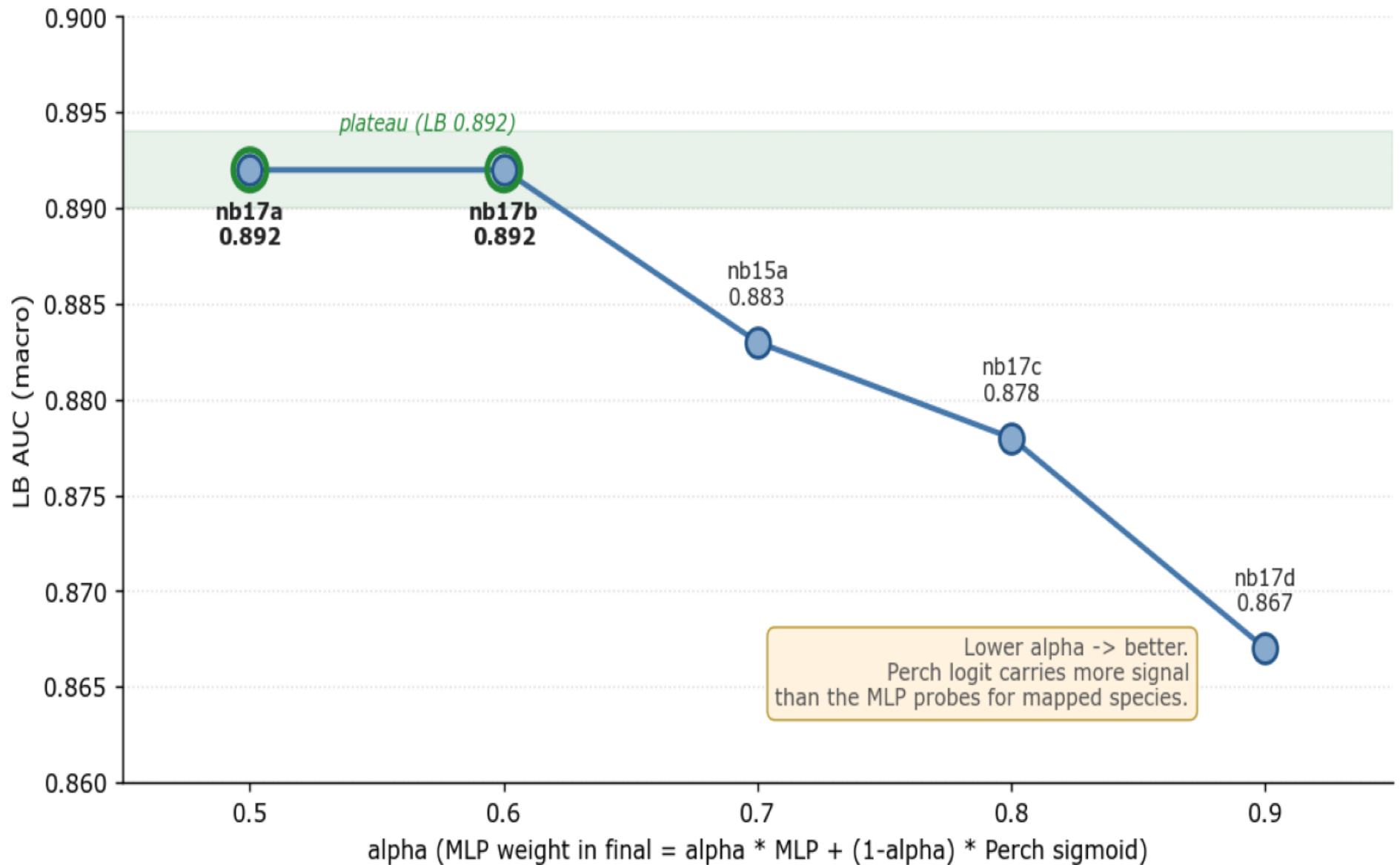
Perch+MLP architecture - frozen Perch v2 features feed 234 per-class probes, blended with Perch's own logit at inference



Frozen Perch v2 features feed 234 per-class probes via torch.bmm; $\alpha=0.6$ blends the MLP correction with Perch's sigmoid at inference time.

Alpha sweep -- the architectural reframe of nb17

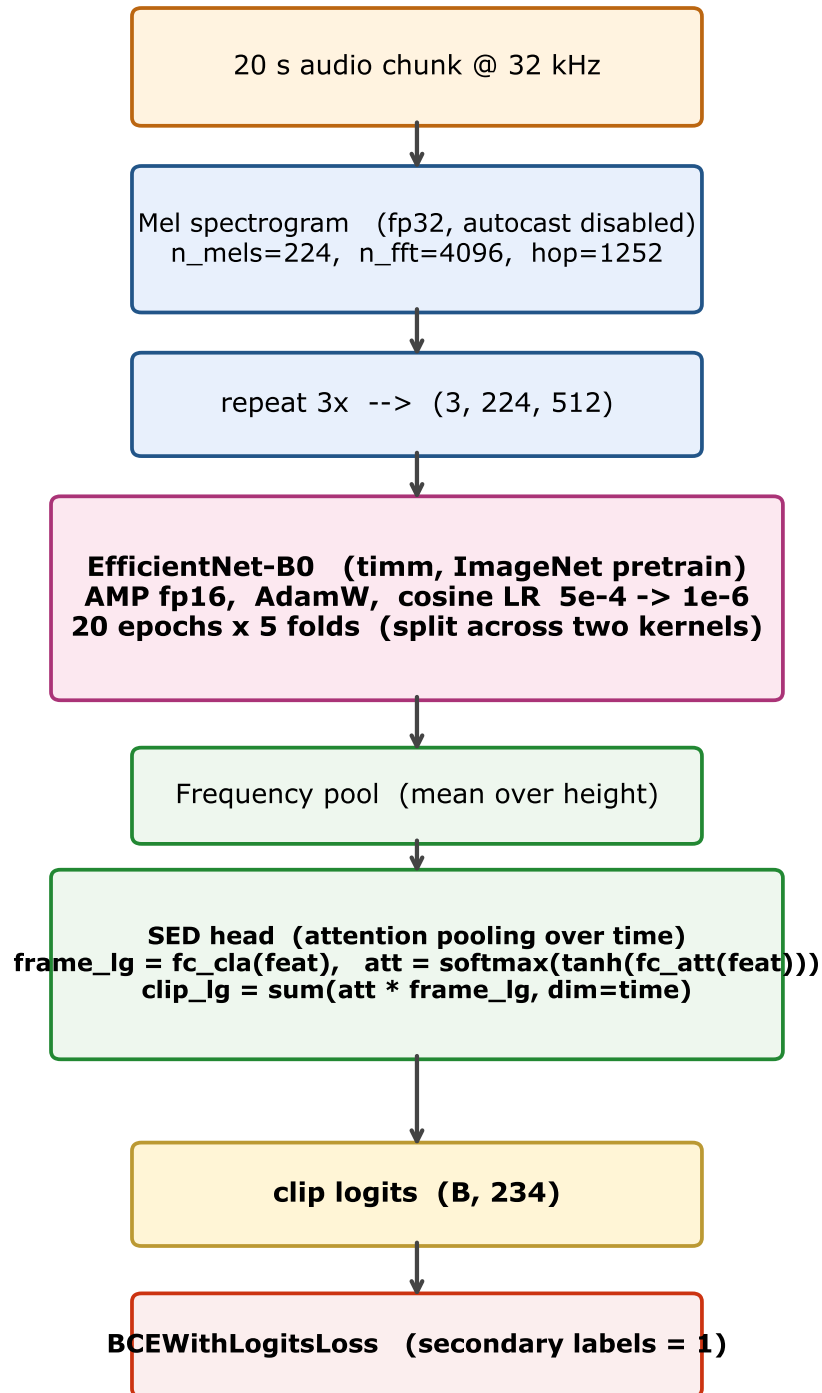
nb17 alpha sweep - Perch logit dominates the blend



Lower alpha -> better LB. Perch's logit carries more signal than the MLP probes for mapped species, so the MLP is a correction layer on top of Perch, not an independent predictor.

CNN+SED architecture (nb20)

*Raw-audio MixUp
($p=0.5$, weight=0.5,
label union)*



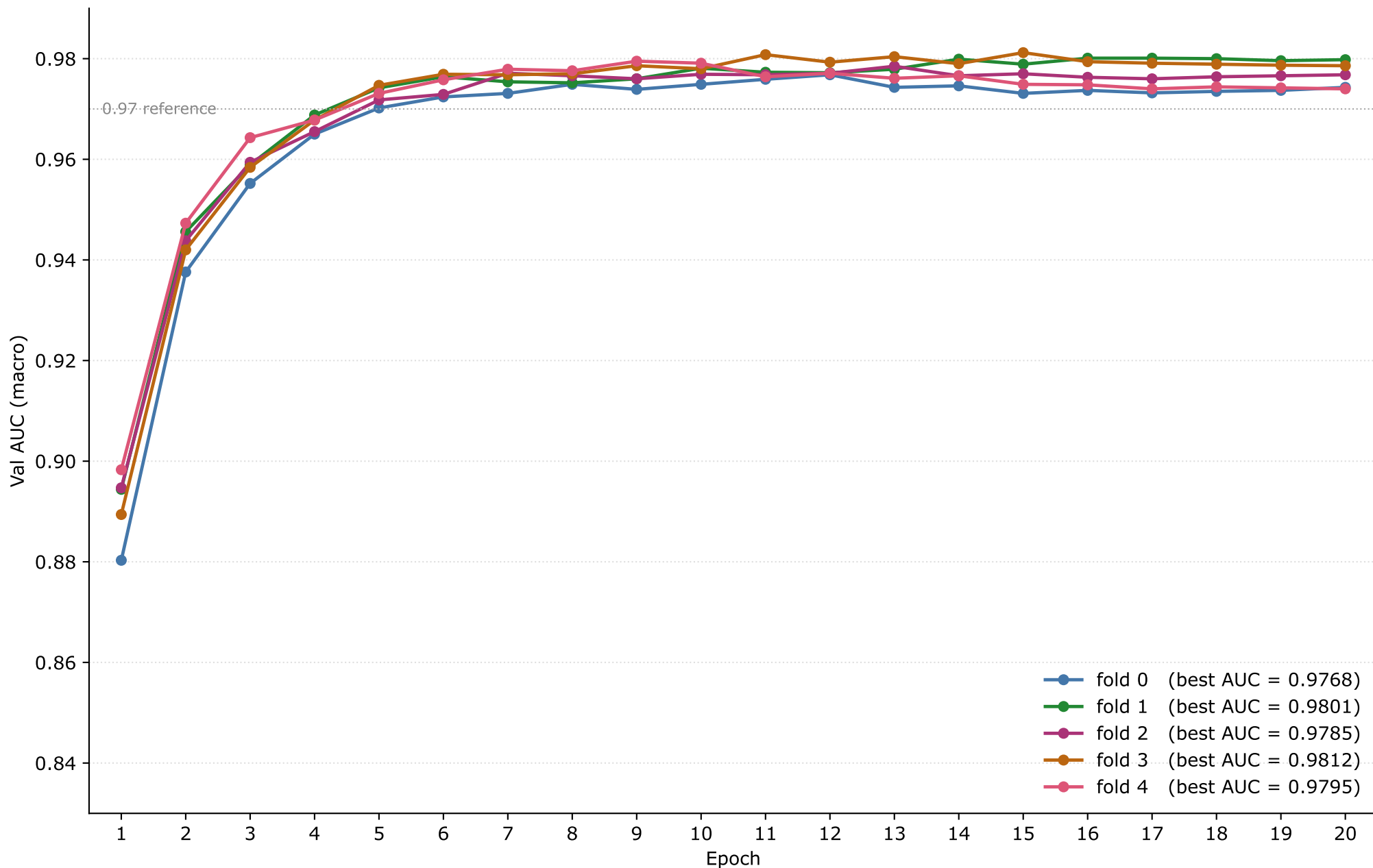
Inference (nb20b)

60 s soundscape ->
9 x 20 s windows
with 5 s stride.
5-fold ensemble.
framewise overlap
averaging into
12 x 5 s output
slots.

+ Top-K postproc
(per-file).

Internet OFF
(timm pretrained
= False)

nb20a training curves -- 5 folds, 20 epochs each



*All 5 folds clear val_auc 0.97 by ~epoch 5 and plateau near 0.978-0.981.
Tight spread (0.0044) signals a stable ensemble; OOF mean = 0.9792.
Caveat: StratifiedKFold-by-primary_label likely overstates the LB.*

nb20a -- 5-fold OOF AUC (split-run training)

Fold	OOF AUC	Best epoch	Source
0	0.9768	12	run 1
1	0.9801	16	run 1
2	0.9785	-	run 2
3	0.9812	-	run 2
4	0.9795	-	run 2
Mean	0.9792	-	5-fold

Run 1 (alexycactus/birdclef-2026-cnn-sed-b0-run1):
trained folds 0+1 in ~6.4 h on T4,
saved fold checkpoints + val_preds to /kaggle/working/.

Run 2 (alexycactus/birdclef-2026-cnn-sed-b0-run2):
mounted run 1 via kernel_sources, trained folds 2+3+4 in ~9.7 h,
ran final 5-fold ensemble inference (staging fallback).

Inference kernel (alexycactus/birdclef-2026-cnn-infer):
internet OFF, 12.2 s for 5-fold ensemble + top-K postproc.
Ready to submit for LB.

nb20a debug chain -- 4 failures before clean training

Version	Failure mode	Fix
v1	Slug 409 conflict	title '0.5' -> slug '0-5' didn't match id
v2	BCELoss autocast-unsafe	swap to BCEWithLogitsLoss
v3	CUDA assert input in [0,1]	NaN slips past .clamp(); add nan_to_num
v4	Collate non-resizable	np.tile slice is a view; np.array(copy=True)
v5 ERR	All train=nan from ep 1	fp16 FFT n_fft=4096 -> autocast(enabled=False)
v5 OK	Healthy ep1 val_auc=0.854	fix verified; split-run launched

All four failure modes added to MEMORY.md as recurring Kaggle audio-CNN gotchas to apply in nb20c / nb21 / nb22 onwards.

Roadmap -- next experiments, sorted by expected LB lever

Notebook	What	Unblocked by	Expected LB
nb20-infer	Submit 5-fold CNN ensemble for LB	ready now	+0 / +0.03 vs Perch+MLP
nb21	Perch+MLP x CNN ensemble (rank avg or weighted)	needs nb20 LB	+0.010 / +0.025
nb20c	Noisy Student round 1 on the CNN	biggest lever	+0.020 / +0.040
nb20c x 4	Noisy Student rounds 2-4 (multi-iterative)	after round 1	cumulative +0.03
nb20d	Backbone diversification (EffNetV2-S, NFNet-L0)	for ensemble div	+0.005 / +0.015 in ensemble
nb22	Top-K post-processing variants	fast iteration	+0.005 (already in nb20-infer)
nb25 (def.)	Xeno-Canto pretraining	high setup cost	+0.025 (2nd place 2025)
nb27	Specialised Amphibia/Insecta model	small but quick	+0.002 / +0.003

Full backlog: [docs/plans/2026-05-14-post-nb20-experiments-backlog.md](#)